List of publications

Shilpi Sengupta

I have authored 19 peer-reviewed research articles, including 15 as first author or co–first author, published in high-impact international journals such as Advanced Energy Materials, Small, Chemical Engineering Journal, ACS Applied Energy Materials, Energy Technology, and Journal of Colloid and Interface Science. My publications primarily focus on the design and development of advanced electrode and electrolyte materials for next-generation lithium- and sodium-ion batteries and solid-state/flexible supercapacitors. These works integrate experimental synthesis, and electrochemical evaluation to address challenges in energy density, flexibility, and electrochemical stability. Also, the full publication in Google scholar.

1. Ternary Metal Sulfides as Electrode Materials for Na/K-Ion Batteries and Electrochemical Supercapacitor: Advances/Challenges and Prospects. A. Pramanik, S. Sengupta et al.. *Advanced Energy Materials, 2024, 14, 2401657.* DOI: 10.1002/aenm.202401657

2. A review on overcoming challenges and pioneering advances: MXene-based materials for energy storage applications. S. Jangra, B. Kumar, J. Sharma, S. Sengupta et al.. *Journal of Energy Storage, 2024, 101, 113810.* DOI: 10.1016/j.est.2024.113810

3. 3D ZnO hexagonal prism-decorated 2D MXene-based high-performance flexible symmetric supercapacitor. S. Jangra, S. Sengupta et al.. *Journal of Energy Storage, 2025, 120, 116366.* DOI: 10.1016/j.est.2025.116366

4. All-Solid-State Flexible Asymmetric Supercapacitors Based on WS_2/rGO/CNT Hybrid Electrodes and Polymer-Based Ionic Liquid Electrolytes. S. Sengupta, M. Kundu. *ACS Applied Energy Materials, 2024, 7, 4243–4251.* DOI: 10.1021/acsaem.4c00670

5. rGO-Wrapped Hexagonal WO_3 Nanorod as Advanced Electrode Material for Asymmetric Electrochemical Supercapacitors. S. Sengupta, M. Kundu. *Energy Technology, 2023, 11, 2300078.* DOI: 10.1002/ente.202300078

6. Experimental and theoretical insights into the supercapacitive performance of interconnected WS_2 nanosheets. S. Sengupta, S. Peters, T. Sadhukhan, M. Kundu. *Physical Chemistry Chemical Physics, 2024, 26, 10301–10309.* DOI: 10.1039/D4CP00206G

7. Self-Assembled 2D WS2 Interconnected Nanosheets: An Anode Material with Outstanding Lithium-Storage Performance. S. Sengupta, M. Kundu. *Energy Technology, 2022, 10, 2200117.* DOI: 10.1002/ente.202200117

8. Ni-Co-A (A = O, S, P) nanosheets for all-solid-state hybrid supercapacitors: Anion substitution unlocks high energy and power density. R. Arulraj, S. Sengupta, M. Kundu. *Chemical Engineering Journal, 2025, 507, 160504.* DOI: 10.1016/j.cej.2025.160504

9. Ternary composites of poly(vinylidene fluoride-co-hexafluoropropylene) with silver nanowires and titanium dioxide nanoparticles as separator membranes for lithium-ion batteries. S. Sengupta, C. R. Tubio, R. S. Pinto et al.. *Journal of Colloid and Interface Science, 2024, 668, 25–36.* DOI: 10.1016/j.jcis.2024.04.149

10. Carbon Free Nanostructured Plate like WS_2 with Excellent Lithium Storage Properties. S. Sengupta, M. Kundu. *ChemistrySelect, 2020, 5, 14183–14189.* DOI: 10.1002/slct.202003510

11. High-performance flexible planner device featuring WS_2 electrode and non-aqueous polymeric ionic electrolytes incorporated with ionic liquid and dual redox additives. S. Sengupta, M. Kundu. *Chemical Engineering Journal, 2024, 490, 151866.* DOI: 10.1016/j.cej.2024.151866

12. Engineering $CoNi_2S_4/Ni_3S_2$ heterostructures: A joint experimental-theoretical study for high-performance solid-state supercapacitors. S. Sengupta, S. Reza, R. Arulraj et al.. *Chemical Engineering Journal, 2025, 507, 167421.* DOI: 10.1016/j.cej.2025.167421

13. Deciphering Sodium-Ion Storage: 2D-Sulfide versus Oxide Through Experimental and Computational Analyses. S. Sengupta, A. Pramanik, C. C. de Oliveira et al.. *Small, 2024, 20, 2403321.* DOI: 10.1002/smll.202403321

14. Carbon spheres decorated–graphene oxide framework as an excellent active material for redox flow battery and supercapacitors. A. Ramadas, A. Dinesh, S. Sengupta et al.. *Environmental Technology, 2023, 44, 2818–2829.* DOI: 10.1080/09593330.2022.2044920

15. Nanostructured MnO_2/CeO_2 composite as anode material for high performance Li-ion battery. S. Sengupta, M. S. Anantha, H. B. Muralidhara, M. Kundu. *Materials Letters, 2022, 308, 131298.* DOI: 10.1016/j.matlet.2021.131298

16. 2H-1T WS_2/rGO/MWCNT Nanocomposites for High-Performance Supercapacitors. S. Sengupta, M. Kundu. *ACS Applied Nano Materials, 2025, 8, 15452–15462.* DOI: 10.1021/acsanm.5c01966

17. 3D-engineered WO_3 microspheres assembled by 2D nanosheets with superior sodium storage capacity. S. Sengupta, C. Sudakar, M. Kundu. *RSC Advances, 2024, 14, 15706–15712.* DOI: 10.1039/D4RA01800A

18. Sustainable WO_3/rGO Nanocomposite Anode for Room and High-Temperature Sodium-Ion Storage. S. Sengupta, A. Pramanik, C. C. de Oliveira et al.. *Small, 2025, 21, e05608.* DOI: 10.1002/smll.202505608

19. Phase-Engineered 1T/2H-WS_2 Heterostructures in Hybrid Carbon Networks for High-Rate and Durable Sodium-Ion Storage. A. Pramanik, S. Sengupta, J. G. Quispe et al.. *Advanced Energy Materials, 2026, e71453.* DOI: 10.1002/aenm.71453